Quotient and Product Topologies: Universal Properties and Gluing

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§1.1 Gluing Endpoints by Quotients

Start with $I = [0, 1]$ and identify $0 \sim 1$. The map

$$q : I \longrightarrow S^1, \qquad q(t) = e^{2\pi i t}$$

is continuous, surjective, and identifies exactly the two endpoints. It therefore induces a continuous bijection $I/(0 \sim 1) \to S^1$. The quotient is compact and the circle is Hausdorff, so the induced bijection is a homeomorphism. This last compact-to-Hausdorff argument is the rigorous step behind the familiar picture of bending an interval into a circle.

The general definition for a surjection $q : X \to Y$ is

$$U \subseteq Y \text{ is open} \quad \Longleftrightarrow \quad q^{-1}(U) \subseteq X \text{ is open.}$$

This is the finest topology on Y for which q is continuous. A map $f : Y \to Z$ is continuous precisely when $f \circ q : X \to Z$ is continuous. Both parts follow immediately by testing preimages of open sets in Z.

A subset $A \subseteq X$ is called *saturated* when

$$A = q^{-1}(q(A)).$$

Thus the open subsets of the quotient correspond to the saturated open subsets upstairs. Near the point obtained from identifying 0 and 1, a neighborhood pulls back to the union of small half-intervals near *both* ends. Keeping only one end is not saturated and therefore does not describe a quotient neighborhood.

The presentation and notation follow the quotient-space and product-space chapters in Munkres' *Topology*, with the universal-property formulation also given in Hatcher's *Algebraic Topology*, Section 0.[1]

§1.2 Collapsing a boundary and reading a glued surface

The quotient D^n/S^{n-1} collapses the boundary of the closed n-disk to a single point. The resulting space is homeomorphic to S^n: the disk interior maps onto a sphere minus one point, while its boundary maps to the missing point. This can be described by stereographic projection, or by regarding the disk as the one-point compactification of its interior $\mathbb{R}^n$. The map is a quotient map because the compact disk maps onto the Hausdorff sphere.

A square provides another family of examples. Identifying one pair of opposite edges with matching directions gives a cylinder; identifying the other pair in the same direction then gives a torus. Reversing one gluing leads to a Klein bottle, whereas a single twisted

[1] See A. Hatcher, *Algebraic Topology*, Section 0, freely available at https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

identification on a strip gives a Möbius band. It is essential to specify *which* edges and the direction of their identifications; a sketch with unlabeled arrows is not enough.

The same gluing can be encoded by an attaching word. For a torus it is

$$aba^{-1}b^{-1},$$

while a Klein bottle can be presented by

$$aba^{-1}b.$$

For the torus, a square is attached as a two-cell to a wedge of two circles. Its boundary loop is a commutator, yielding

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

Thus the arrow labels in the square are the start of a genuine algebraic calculation of a fundamental group.

§1.3 Products use the opposite continuity test

For topological spaces X, Y, the product topology on $X \times Y$ is generated by the rectangles

$$U \times V, \qquad U \subseteq X,\ V \subseteq Y \text{ open}.$$

Indeed,

$$U \times V = \pi_X^{-1}(U) \cap \pi_Y^{-1}(V),$$

so it is the coarsest topology making both projections π_X, π_Y continuous. Its mapping property reads

$$h : Z \to X \times Y \text{ continuous} \quad \Longleftrightarrow \quad \pi_X \circ h,\ \pi_Y \circ h \text{ continuous}.$$

Compare this with the quotient criterion, where the universal test concerns *maps out of* the quotient. The two constructions are both specified by functions, but by different directions of functions:

construction	canonical maps	continuity checked by
product	$X \times Y \to X, Y$	component maps into the product
quotient	$X \to X/\sim$	maps defined on the quotient

A continuous map into a product requires coordinatewise continuity; a continuous map out of a quotient can be checked before identifying the points.

§1.4 When quotients behave badly

Let $q : \mathbb{R} \to Y$ identify all rational numbers as one point and all irrational numbers as another. Neither $\mathbb{Q}$ nor its complement is open in $\mathbb{R}$, and neither has open nonempty saturated proper subsets. Therefore Y carries the *indiscrete* two-point topology. It cannot be Hausdorff. This is a concrete warning that quotient spaces of Hausdorff spaces need not remain Hausdorff.

There is a second pitfall in mixing the two constructions. Given quotient maps

$$q : X \to X/\sim, \qquad r : Y \to Y/\approx,$$

the product $q \times r : X \times Y \to (X/\!\sim) \times (Y/\!\approx)$ is a continuous surjection, but it need not be a quotient map for arbitrary spaces. The induced set-theoretic bijection

$$(X \times Y)/(\sim \times \approx) \longrightarrow (X/\!\sim) \times (Y/\!\approx)$$

may therefore fail to be a homeomorphism. If both q and r are *open quotient maps*, however, their product is open and surjective, hence a quotient map; the displayed bijection is then a homeomorphism. The endpoint-identification map above is closed rather than necessarily open, but compactness/Hausdorffness supplies an alternative direct proof for the torus.

The point of the universal-property approach is that it forces these hypotheses into view. A drawing can correctly predict a circle or torus while hiding the conditions under which its proposed quotient carries the intended topology.